Blessed G/Michael Catholic General Secondary School
Mathematics Test one 2018 E. C for grade 11th

Time allowed: 1:30

I. Write true if statement is correct and if it is incorrect say false

True 1. If n(A) = p, n(B) = q then the n(AxB) = pq. For relation A to B.

False 2. The relation R = {(x,y) : $y = \sqrt{1-x}$, The domain of R is $[0, \infty)$

False 3. The solution set of $\frac{-1}{x^2+1} > 0$ is $(1, \infty)$

True 4. For two functions f and g, the domain of gof is the set of all x in the domain of f.

True 5. For x, y $\in IR$, /x+y/ ≤ /x/ + /y/.

II. Choose the best answer among the given alternatives.

C 1. Which one of the following is true about the relation R = {(x,y) : $y = \sqrt{x}$ }

A. Domain of R = IR
B. Range of R = $(0,\infty)$
C. Range of $R^{-1} = [0, \infty)$
D. Domain of R^{-1} = IR

A 2. Let A = {1,2,3,5} and B = {4,6}, then which one of the following define a relation R from B to A:

A. {(4,2), (4,3) (6,3), (4,1)}
B. {(2,6), (3,6) (4,5)}
C. {(6,1) (6,5) (3,6)}
D. {(4,3), (4,2), (1,6)}

D 3. If A = {2, 4,6,9} and B = {4,6,18,27,54}, a∈ $A, b \in B$ the set of order pairs such that a is a factor of b and a<b is ----------.

A. {(2,4) (2,18), (6,18) (9,18), (6,4)}
B. {(2,4) (6,4) (6,6)}
C. {(2,18) (6,4), (9,6)}
D. {(2,4) (2,18) (6,18)}

D 4. Let R = {(x,y) : $y \geq x^2$-16 and y ≤ 9} which one of the following is true about R.

A. Domain of R = [-4,4]
B. Domain of R^{-1} = [-16, 9]
C. Range of $R^{-1} = [-5, \infty)$
D. Range of R = [-16, 9]

B 5. Which one of the following is not a function?

A. R = {(x,y) : $y = \sqrt{x-1}$}
B. R = {(x,y) : $x^2 + y^2 = 4$}
C. R = {(x, y) :$y = x^2$-4}
D. R = {(x,y) : $y = |x-1|$

C 6. Which one of the following is true about $f(x) = x^{4/3}$

A. Domain of f is IR/ {0}
B. Range of f is IR/{0}
C. Symmetric w.r.t y- axis
D. Range of f is $[1, \infty)$

A --7. Let $f(x) = x^{-m/n}$, for an odd natural number 'm' and an even natural number 'n', then which one of the following is false about f:

A. Domain of f is $[0, \infty)$
B. Range of f is $(0,\infty)$
C. f has no symmetry.
D. for the graph $x \to 0, y \to \infty$

C --8. The solution set of $/x+1/ = 0$ is ----------,

A. 1 B. $\emptyset$ C. -1 D. 4

C --9. Which one of the following is true about $f(x) = \left|\frac{1}{x+1}\right|$.

A. Domain is IR $-\mathbb{R}\setminus\{-1\}$
B. Range of f is $(2, \infty)$
C. Symmetry of to the line $x = -1$
D. Range of f is IR

A --10. The domain of $f(x) = |x|+2$ is ----------.

A. IR B. $[0, \infty)$ C. $(0, \infty)$ D. $[2, \infty)$

B --11. Which one of the following is true?

A. $x\,\text{sgn}\,x = \sqrt{x}$
B. $x\,\text{sgn}\,x = /x/$
C. The domain of $x^2\text{sgn}x$ is $[0, \infty)$
D. Range of sgnx is IR

C --12. Let $f(x) = [x]$, then which one of the following is true.

A. $[x+y] = [x] + [y]$
B. $[x] - [y] = [x - y]$
C. $[x] + [y] \leq [x+y]$
D. $[x]^2 = [x^2]$

A --13. The solution set of $[x-4] = 6$ is ----------.

A. [10,11) B. (10,11) C. (4,6] D. [10,14)

B --14. Let $f(x) = 2x$[illegible], then what is the value of g(x) for $f(g(x)) = 6x^2+4x+2$.

A. $3x^2+4x+4$ B. $3x^2+2x+1$ C. $2x^2-4x+1$ D. x^2-4x+6

B --15. Which one of the following is not inverse of each other?

A. $f(x) = x^3$, $g(x) = \sqrt[3]{x}$
B. $f(x) = x^2$, $g(x) = \sqrt{x}$
C. $f(x) = 8x$, $g(x) = \frac{x}{8}$
D. $f(x) = \frac{2}{3}x + 2$, $g(x) = \frac{3}{2}x + 2 = \frac{3}{2}x - 3$

b --16. The domain of $\frac{x^2-9}{x^2-7x+10}$ is ----------.

A. IR B. IR/{ 2,5} C. $[0, \infty)$ D. $[1, \infty)$

1. List the major cereal and pulse crops

C 17. The simplified form of $\frac{x^4+18x^2+81}{x^2+9}$ is

A. x+3 B. x^2+4 C. x^2+9 D. $\frac{1}{x+4}$

A 18. Which one of the following is the decomposition form of $\frac{2+6x}{x^2+6x+5}$ is ---------.

A. $\frac{7}{x+5}-\frac{1}{x+1}$ B. $\frac{7}{x+5}+\frac{2}{x+1}$ C. $\frac{7}{x+5}-\frac{2}{x+1}$ D. $\frac{1}{x+4}$

B 19. The solution set of $\frac{x^2+1}{x^2+x-2}<0$ is

A. $(-\infty,2)\,u(1,3]$ B. $(-2,1)$ C. $[-2,\infty)$ D. IR

D 20. Which one of the following is true about the graph f(x) = $\frac{x-1}{x^3-3x^2+4}$ $-1-3(1)+4$

A. It has an obligue asymptote ✗
B. Its vertical asymptote is at x = 1 ✗
C. Its horizontal asymptote is at y = 1 ✗
D. Its vertical asymptote is at x = -1

III. Short Answer

1. Find the inverse of f(x) = $\frac{2x+4}{2x-1}$ $\frac{x+4}{2x-2}$
2. Perform and simplify $\frac{x^2-x-6}{3x^2-12} \div \frac{x^2-3x}{2-x}$ is ---------- $\frac{1}{-3x}$
3. The decomposition form of $\frac{6x^2-14x-27}{x^3-4x^2-3x+18}$ is ---------- $\frac{3}{x+2} - \frac{5}{x-3} + \frac{4}{(x-3)^2}$
4. A tree casts a shadow of 34 feet at the time when a3 feet tall child casts a shadow of 1.7 feet. What is the height of the tree? ~~feet~~ 60 feet
5. Let f(x) = $\frac{x^3+2x+1}{x^2+1}$, then find the horizontal asymptable? It has no horizontal asymptote

B = 5
A = 1
C = -3

~~28~~

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True/False

1. True
2. False
3. False
4. True
5. True

Choosing

1.	C	6.	C	11	B	16	B		
2.	A	7.	A	12	C	17	C		
3.	D	8.	C	13	A	18	A		
4.	D	9.	C	14	B	19	B		
5	B	10.	A	15	B	20	D		

Short answer

1. $\frac{x+4}{2x-2}$ for $x \neq 1$

2. $\frac{1}{-3x}$ for $x \neq 0$

3. $\frac{1}{x+2} + \frac{5}{x-3} - \frac{3}{(x-3)^2}$ for $x = R \setminus \{-2, 3\}$

4. 60 feet

5. It has no horizontal asymptote

30/30

Excellent

Keep it up!!